

SIMATS ENGINEERING
ASSESSMENT TOOL 3 – VIVA QUESTIONS

Course:ETHICAL HACKING

Code:ITA1407

CO: CO5

Duration:29-08-2026

Total Marks: 100

Reg No / Name :192511230

COURSE OUTCOME COVERED

CO5: *Analyze vulnerabilities in Windows and Linux systems and evaluate security weaknesses in messaging, web, and mobile applications using appropriate exploitation techniques and hacking tools. (BT5)*

Code of Conduct:

I Aluri Poojitha_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Instructions: Answer all questions. Each question carries 4 marks.

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1. **How would you evaluate the security posture of a Windows system after identifying multiple open services?**
 - Identify all open ports and the services running on them.
 - Check service versions for known vulnerabilities.
 - Verify whether the services are necessary and securely configured.
 - Check firewall rules, patches, authentication, and access controls.
 - Prioritize high-risk services and recommend remediation.
 2. **Which factors would you consider when choosing a hacking tool for Windows vulnerability assessment, and why?**
 - Accuracy – should identify vulnerabilities correctly.
 - Reliability – should produce consistent results.
 - Compatibility – must support the Windows version and services.
 - Safety – should minimize system disruption.
 - Reporting – should provide clear evidence and reports.
 3. **How would you evaluate the security risk associated with an exposed RPC service?**
 - Whether RPC is accessible from untrusted networks.
 - The RPC service/version and its known vulnerabilities.
 - Authentication and access-control settings.
 - Whether unnecessary RPC endpoints are exposed.
 - The possible impact, such as unauthorized access or privilege escalation.

4. **Why are Windows Null Sessions considered a security weakness, and how would you assess their impact? Check whether anonymous connections are allowed.**
 - Determine what information can be obtained.
 - Check whether usernames, shares, or other sensitive information are exposed.
 - Assess whether the information could help further attacks.
 - Disable unnecessary anonymous access
5. **A Windows shared folder allows unauthorized users to access sensitive files. How would you evaluate the severity of this vulnerability? What sensitive information is accessible.**
 - Which users or groups have access.
 - Whether users can only read or also modify/delete files.
 - Whether the share is accessible remotely.
 - The business impact of the exposed information.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Mark
Understanding of Problem Context	20	Clearly understands the security problem and accurately identifies vulnerabilities, risks, and requirements	Good understanding of the security problem with minor gaps	Basic understanding; some security requirements or vulnerabilities are unclear	Poor understanding or misinterpretation of the security problem	
Application of Concepts	30	Accurately applies appropriate ethical hacking concepts, tools, and exploitation techniques to analyze complex security problems	Applies relevant security concepts and tools correctly with minor errors	Applies basic concepts with limited accuracy or tool selection	Fails to apply appropriate security concepts or tools	
Problem-Solving Approach	20	Demonstrates a logical, well-structured, systematic, and efficient approach to identifying and evaluating security weaknesses	Mostly logical and systematic approach with minor gaps	Basic approach with some inconsistencies in vulnerability analysis	Illogical, unclear, or inappropriate approach to solving the security problem	

Accuracy of Solution	20	Provides a completely correct security analysis with appropriate justification, evidence, and suitable prevention	Mostly correct analysis with minor errors or gaps	Partially correct analysis with limited justification	Incorrect or incomplete security analysis	
Clarity	10	Provides clear, well-organized, technically sound, and properly explained responses using appropriate security	Understandable response with minor clarity or terminology issues	Limited clarity and explanation; uses technical terms inconsistently	Poorly presented, unclear, and difficult to understand	
Total	100					/100